01. Introduction: PrismSpace Web Platform

PrismSpace is a local-first, **web-based multi-agent intelligence platform** and interactive website workspace that allows users to orchestrate autonomous AI swarms directly from modern web browsers. It converts natural-language user goals into dependency-aware **Directed Acyclic Graphs (DAGs)** for parallel sub-task execution.

- **Modern Web Architecture:** Built as a responsive web application (Next.js 15, React 19) requiring no complex local environment setup.
- **Live Reactive Interface:** Features an interactive visual DAG swarm visualizer with real-time Server-Sent Events (SSE) token and step streaming.
- **Dynamic Multi-Model Routing:** Pairs Groq, OpenAI, and NVIDIA NIM with standardized tool connectivity via Model Context Protocol (MCP).
- **In-Browser Privacy & Safety:** Stores all session state locally in client-side IndexedDB (`PrismDB`) with human-in-the-loop approval gates.

01. Introduction: PrismSpace Web Platform

PrismSpace is an interactive, **web-based multi-agent intelligence platform** that allows users to orchestrate autonomous AI swarms directly from a modern web browser. Designed as a local-first web application, it converts natural-language user goals into dependency-aware **Directed Acyclic Graphs (DAGs)** for parallel sub-task execution. By pairing dynamic multi-model routing (Groq, OpenAI, NVIDIA NIM) with standardized tool access via the **Model Context Protocol (MCP)**, the website coordinates specialist agents for everyday productivity and technical workflows—featuring real-time streaming telemetry, browser-native IndexedDB persistence (`PrismDB`), and in-browser human-in-the-loop safety verification within a unified, reactive web workspace.

01. Introduction: PrismSpace Web Platform

PrismSpace is an interactive, **web-based multi-agent intelligence platform** that enables users to solve complex everyday and technical goals directly inside their web browser. It transforms natural-language requests into dependency-aware **Directed Acyclic Graphs (DAGs)**, orchestrating specialized AI swarms with dynamic multi-model routing, open tool execution, and local-first privacy.

1. Reactive Web Workspace

- **Browser-Based Access:** Responsive web application built with Next.js 15 and React 19.
- **Visual Swarm Monitor:** Interactive web canvas displaying live DAG execution progress.
- **Real-Time Streaming:** Server-Sent Events (SSE) stream agent thoughts, logs, and token outputs.

2. Intelligent Swarm Routing

- **Dynamic DAG Scheduling:** Splits requests into parallel sub-tasks ($K \leq 5$) with dependency barriers.
- **Multi-Model Backends:** Routes sub-tasks to Groq (speed), OpenAI (reasoning), or NVIDIA NIM.
- **Open Tool Protocol (MCP):** Standardized web connectivity to Gmail, Drive, Filesystem, and Web.

3. Local Privacy & Safety

- **Client-Side PrismDB:** ACID database running locally in browser IndexedDB via Dexie.js.
- **Zero Cloud Data Exposure:** Chat histories, files, and tokens remain on the client's machine.
- **In-Browser HITL Gating:** High-risk operations pause for explicit user click approval.